ME325 · MECHANICAL ENGINEERING GROUP PROJECT

Adaptive Stiffness-Based Selective Tea Harvesting Mechanism

Engineering a passive, stiffness-sorting gate so a battery shear can pluck two-leaves-and-a-bud as selectively as a trained hand.

Group 12 - Tea Harvest

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What We Set Out to Build

PRIMARY OBJECTIVE

- **Design a precision tea-harvesting mechanism that selectively harvests shoots meeting the bud + 2–3 leaves plucking standard.**

Study existing systems

Analyse mechanical tea harvesters used in Sri Lanka and identify their operational limitations.

Quantify the failure modes

Evaluate leaf damage, coarse-leaf inclusion, and poor terrain adaptability in non-selective harvesting.

Everything in the machine's cutting path is cut together

Match hand-plucking quality

Improve harvesting efficiency while keeping tea quality comparable to manual plucking.

Lightweight & ergonomic

Develop a <2 kg mechanism suitable for steep, uneven tea-plantation environments.

Milestones & Completion Status

Milestone	Due	Completion Status	Remarks
Literature Review & Problem Identification	Week 2	100%	TRI field study complete; 35–40% yield-loss problem confirmed
Requirement Analysis	Week 3	100%	Design requirements & constraints finalised
Concept Generation	Week 4	100%	Three candidate concepts generated & benchmarked
Concept Evaluation & Selection	Week 5	100%	Stiffness-based sorting gate selected
Material Selection / Engineering Calculations	Week 6, 7	70%	Bending-force field data + statistical threshold derived
3D CAD Modelling / Simulation & Validation	Week 8, 9, 10	70%	Gate geometry modelled; laser-cut prototype field-tested; FEA validation ongoing
Comparative Evaluation	Week 11	20%	Host machine acquired; integration benchmarking starting
Final Design Recommendation	Week 12	0%	Upcoming

Problem Definition

Across three months of field work and engineering analysis, we moved from a confirmed industry problem to a statistically-derived, mechanically-validated sorting threshold.

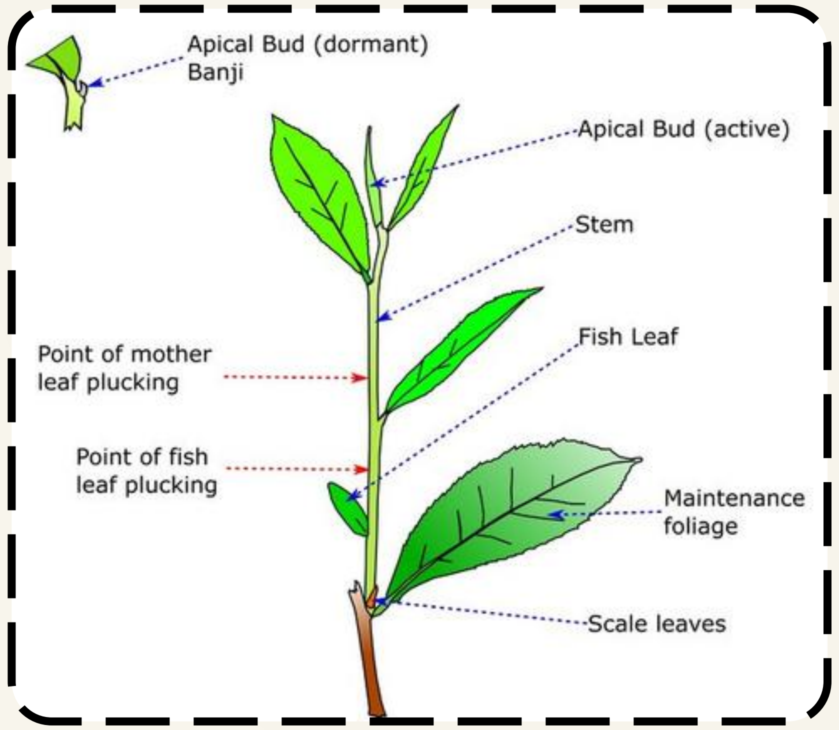
THE PROBLEM

- **TRI Ratnapura Field Insight:** Observation of existing machines confirmed they lack any selectivity—cutting buds, stems, and immature shoots together.
- **Critical Yield Loss:** Non-selective cutting accidentally removes "Arimbu" (immature shoots), destroying the crop intended for the next 7-day plucking round.

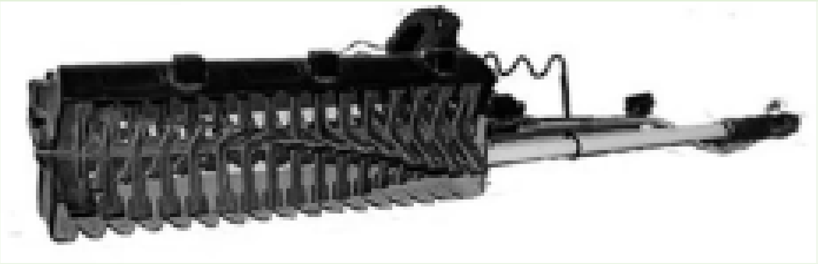
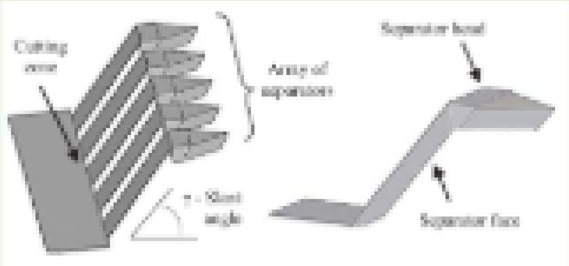
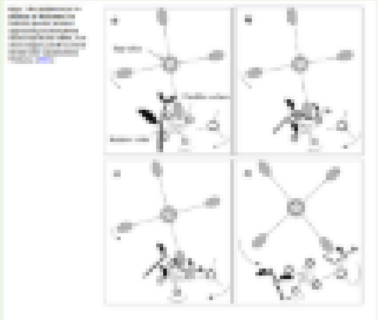
35–40%

estimated yield loss from non-selective mechanical harvesting

- **Permanent Plant Damage** (Non-selective cutting leads to long-term decline in bush health).
- **Industrial Damage:** High inclusion of coarse wood stems disrupts automated factory sifting and lowers the final value grade of the tea.
- **Economic Crisis:** Manual harvesting is highly selective but increasingly unsustainable due to labor shortages and high costs (Rs. 375 per kg)



Existing Selective Systems and Limitations

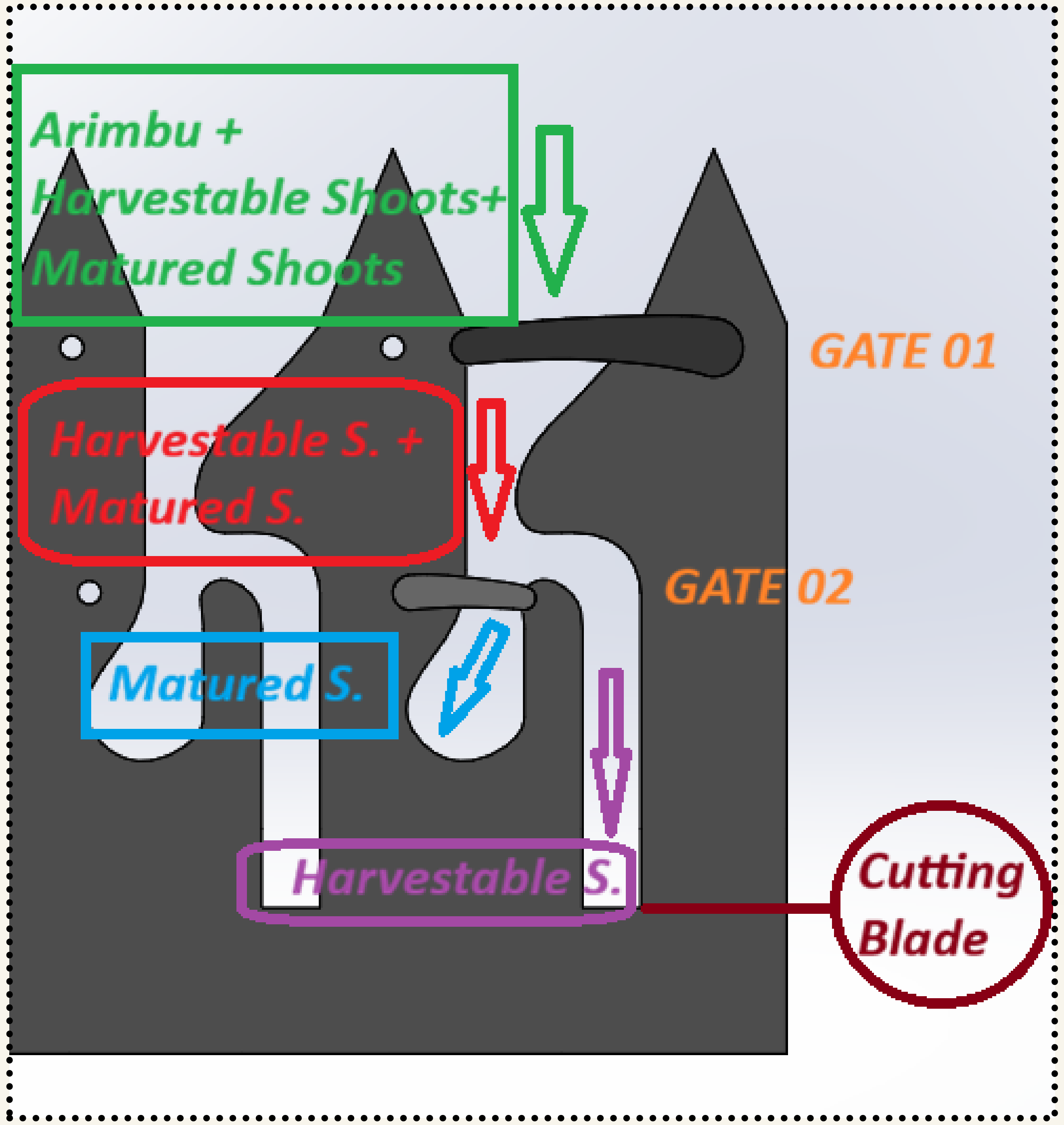
Existing system		Working principle	Limitation
Manual plucking		Experienced workers manually select the "bud + 2 leaves" standard.	High labour cost, slow, worker shortage
TRI selective shear		Hand-operated scissor mechanism providing ~2× manual speed with good selectivity.	Good quality but low productivity, still manual
Battery-powered rotary		Rotating blade chops leaves and pushes them into a collection bag	Poor cutting process; lower production rate due to smaller cutting span; not truly selective
Comb-type Gauging Device		Passive comb separators at 74mm height use shoot stiffness to guide harvestable stalks to the blade.	Shoot "bunching" destroys selectivity; prone to cluttering; only 52% harvestable selection efficiency.
Williames roller selective plucker		Two rollers trap the tea shoot, apply tension, and snap the tender shoot	Roller gap and pressure-point spacing must be accurately controlled; wrong clearance can harvest small/medium shoots or miss shoots

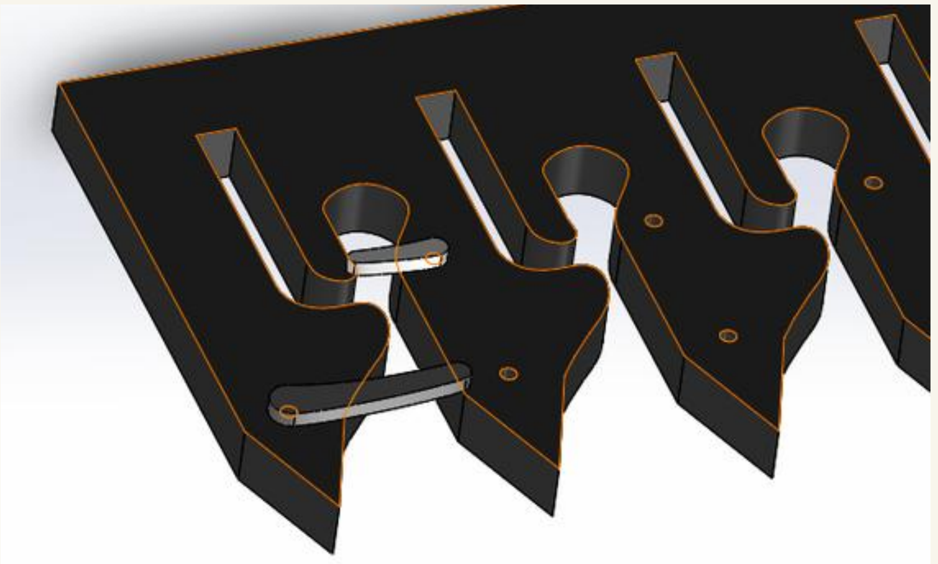
- ***Therefore, our design focuses on a low-cost passive stiffness-based guide that can be attached to an existing battery harvester.***

Concept Selection

BENCHMARKING & DECISION

- **Speed vs. Selectivity:** Reviewed existing selective and non-selective tea harvesting systems to identify the main gap: high speed but poor selectivity.
- **Concept Selection:** Evaluated three options (Sensor-based, Optical, and Stiffness-based)
- **Why Stiffness-Based Gate?** Selected because it is purely mechanical, requires no expensive sensors.
- **Dual-Filter Mechanism:**
  - Gate 1: Flexible Arimbu shoots are filtered out because they lack the force to trigger the gate.
  - Gate 2: Mature shoots apply enough force to open the gate and are diverted into the slot, while harvestable shoots do not trigger the gate and continue toward the blade for cutting.





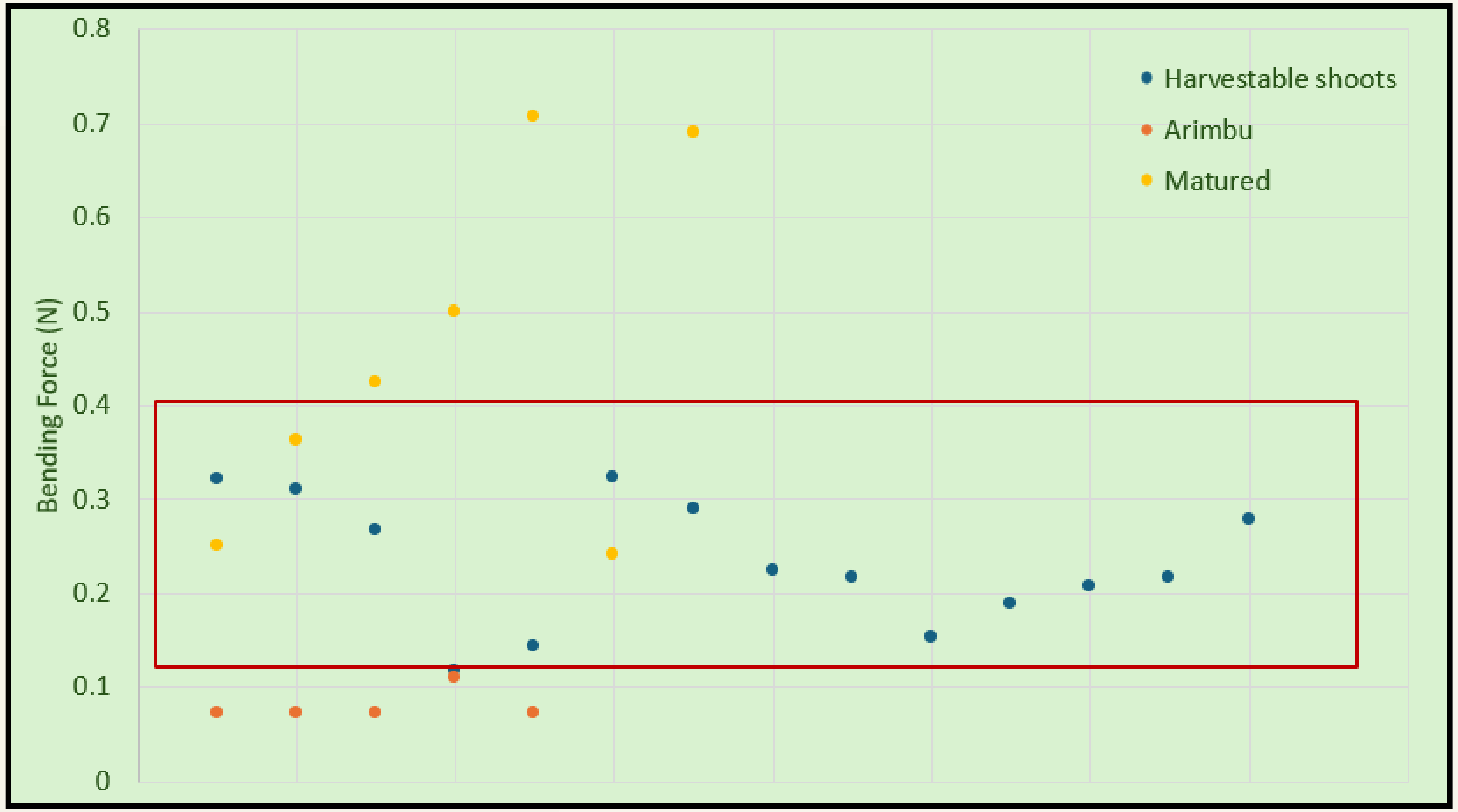
Data Colection

FIELD DATA COLLECTION

- **Location:** Conducted primary data collection at **TRI Thalawakale**
- **Apparatus:** Utilized a custom-built, pulley system to measure bending forces directly on living plants

30 samples

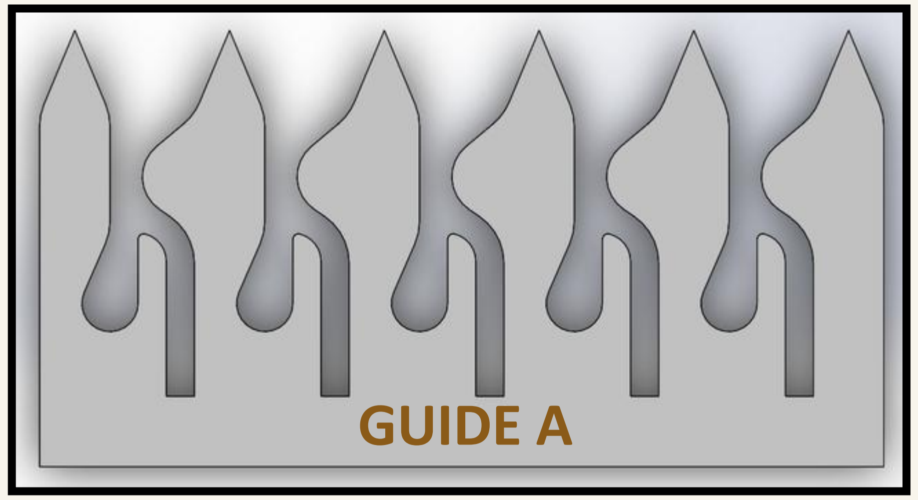
- **Immature (Arimbu):** Very soft, mostly below **0.11 N**.
- **Pluckable Range:** Mainly between **0.12 N** and **0.32 N**.
- **Mature Wood:** Need higher force, usually above **0.50 N**

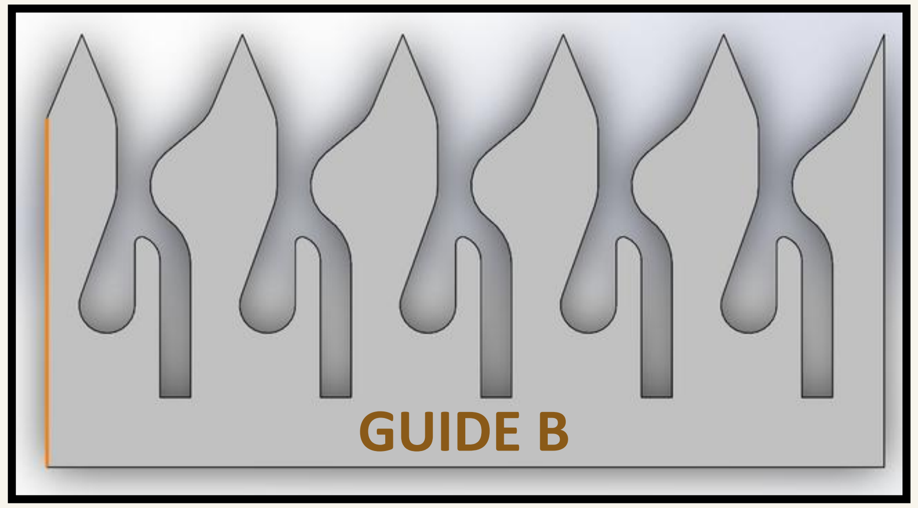


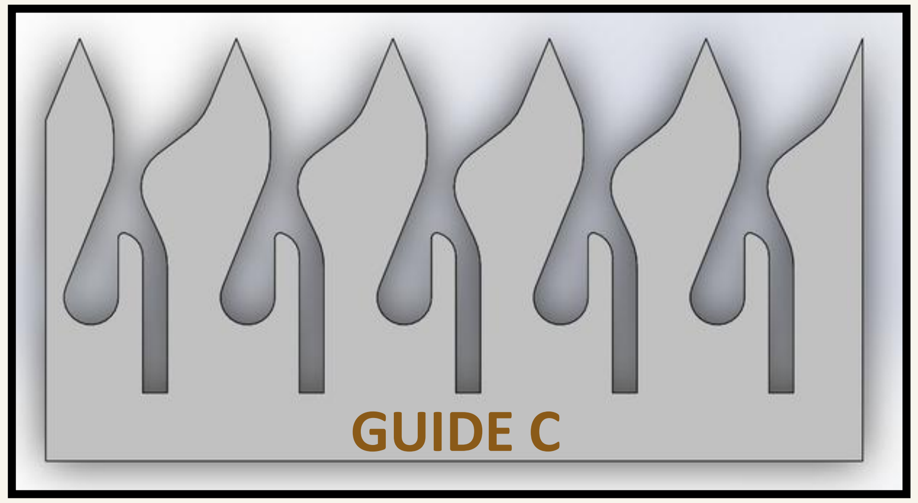
Mechanism Design, Prototyping & Validation

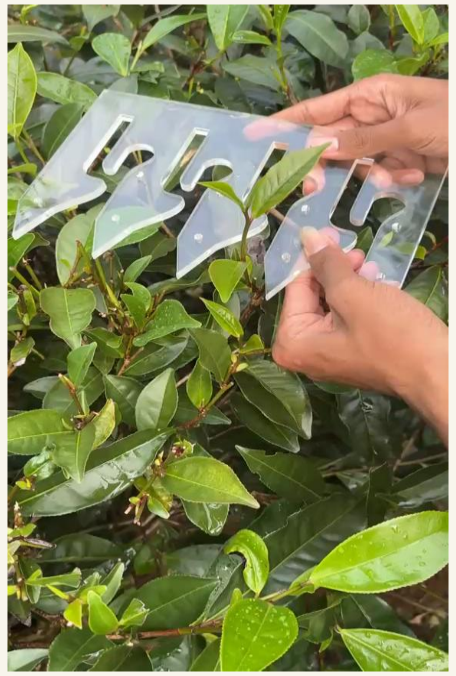
FIELD VALIDATION & ITERATIVE SUCCESS

- **Initial Success (Guide A):** Field trials at Meewathura plantation validated the basic mechanical concept and confirmed smooth sliding clearance for tea shoots.
- **Identified Bottlenecks:**
  - Shoot Escape: Harvestable shoots slipped past the 2nd gate due to angular misalignment.
  - Clustering: Tight bunches of shoots overpowered the gate springs, breaking selectivity.
  - Falling Defect: Severed shoots tumbled onto the ground instead of into the collection bag.
- **Current Status:** We are currently testing Guide B and Guide C in the field









Field Trial of Guide A

Guide B & C

Engineering Calculations & Mathematical Basis

Establishing the scientific foundation for the stiffness-sorting mechanism through statistical analysis and mechanical modeling

Selection Logic: Optimal Threshold Force (Fopt)

The mechanism is calibrated to the intersection of pluckable (*P*) and mature (*M*) force distributions to minimize sorting errors.

- **F / μ / σ:** Bending force, its Mean, and Standard Deviation.
- **Key Result: Fopt=0.34 N** (The mechanical trigger point).

$$\frac{F_{\text{opt}} - \mu_p}{\sigma_p} = \frac{\mu_m - F_{\text{opt}}}{\sigma_m}$$

Mechanical Gate Design

Torsion springs are preloaded to ensure gates only trigger at the correct maturity level.

- **T / θ / d:** Spring torque, Gate angle, and Contact distance from pivot.
- **Fmax:** Marginal maximum force (μ+3σ) to ensure 99.7% filtering accuracy.
- **Requirements: T1 ≥ 0.0027 Nm** (Arimbu Filter) | **T2 ≥ 0.0014 Nm** (Stiffness Filter).

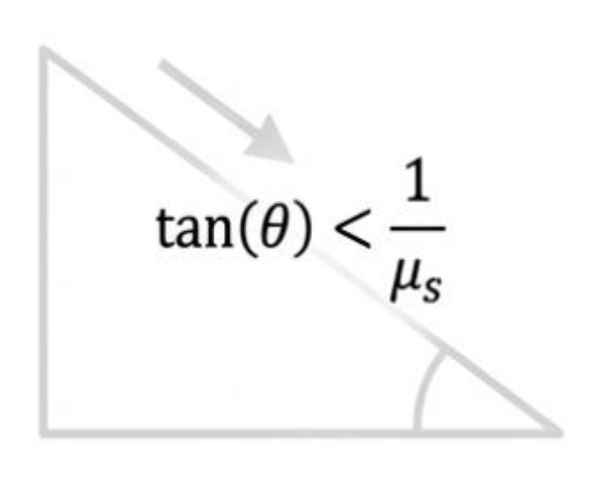
$$T > (F_{\max} \times \sin(\theta)) \times d_{\text{gate}}$$

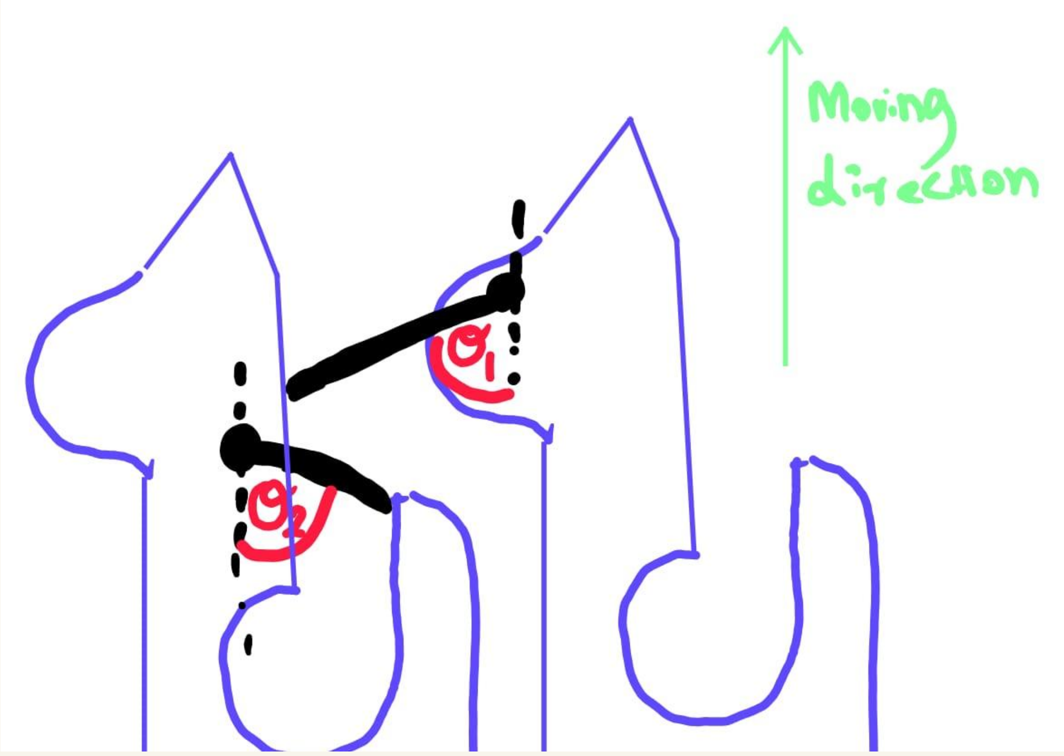
Engineering Calculations & Mathematical Basis (Cont.)

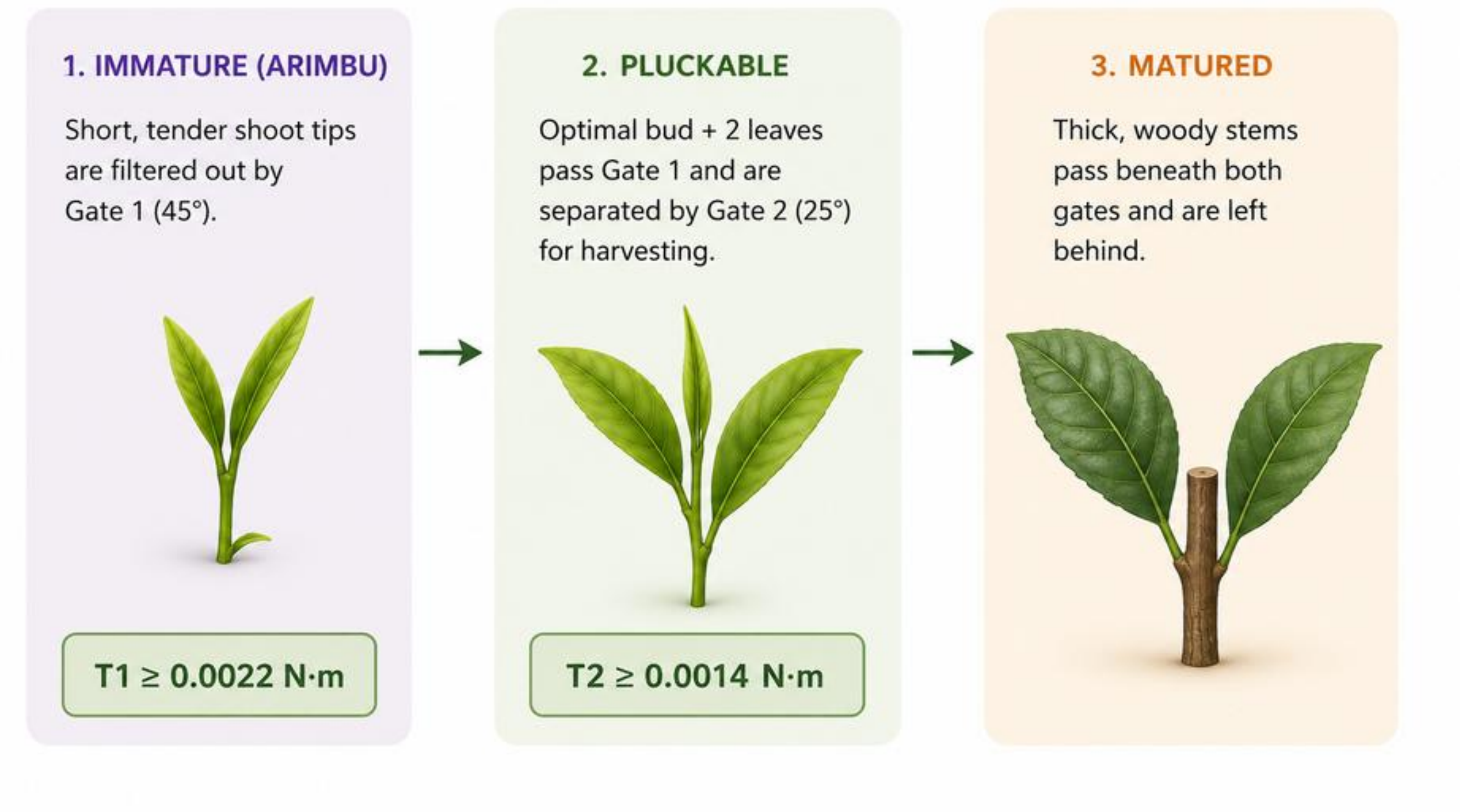
Operational Reliability

Gate angles (ϑ) are designed to prevent jamming based on material friction (μs).

- **Validation:** Designed angles (**45∘** and **25∘**) satisfy the condition for PLA

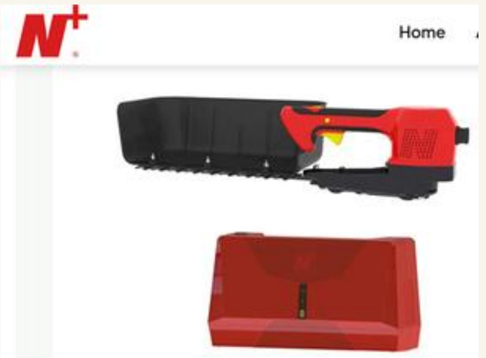






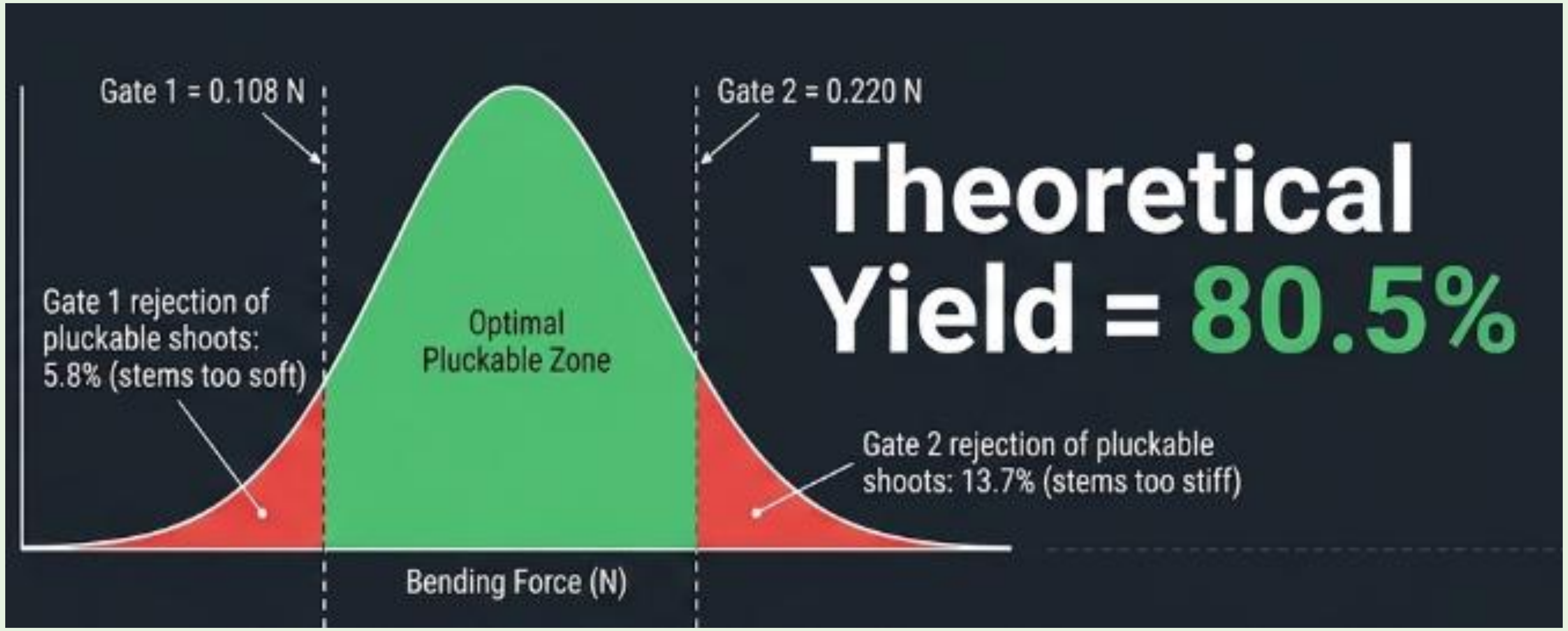
HOST PLATFORM

- Surveyed 6 commercially-available single-hand battery harvesters on motor power, blade, weight, and capacity.



Theoretical Yield & System Efficiency

- Total Theoretical Yield:
  - ❑ **80.5%** Calculated prediction shows that over 80% of ideal pluckable shoots are successfully harvested
- System Rejection Rates:
  - ❑ Gate 1 Rejection (5.8%): Shoots too flexible (< 0.108 N) to trigger the filter
  - ❑ Gate 2 Rejection (13.7%): Shoots too stiff (> 0.220 N), filtered to ensure zero mature wood inclusion
- Statistical Basis:
  - ❑ Derived from TRI Thalawakale field data:
    Mean (μ) = 0.174 N
    Std Dev (σ) = 0.042 N



Integration Plan & Next Steps

WHERE WE STAND

Week 9 of 12 — concept selected, field data collected, and engineering calculations complete. The 3D model is being built and validated; a laser-cut prototype is already field-tested, and integration onto the acquired host machine is underway.

- ***Status Update:*** *We have purchased a battery-operated machine.*
- ***Next Phase:*** *We plan to assemble the selection mechanism (Guide) to it and conduct full-scale performance checks at the Meewathura tea field*

01
Mount the gate stack
Fix the two-gate comb-guide assembly ahead of the acquired shear's reciprocating blade.

02
Calibrate spring preload
Set torsion-spring preload to $T_1 \geq 0.0022$ N·m and $T_2 \geq 0.0014$ N·m using the derived thresholds.

03
Run controlled cutting trials
Test on live bushes .
(Testing on live bushes at the Meewathura tea field)

04
Benchmark vs. manual plucking
Measure selectivity rate, leaf-damage incidence, and throughput against the TRI hand-shear baseline.

Q&A

Thank you — questions & discussion welcome

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GitHub

Project Repository

github.com/TADilhara/Selective-Tea-Harvester-Stiffness-Gate

Scan the QR code